



1. Motivation & Objectives

- **Research Question:** Is modern global warming driven by anthropogenic CO₂ or natural forcings (solar TSI & volcanic activity)?
- **From-Scratch Record:** Reconstruct surface record from raw land/sea grids (1854–2025).
- **Autocorrelation Guardrails:** Prevent false significance (p -hacking) using Bretherton N_{eff} corrections.

2. Observational Datasets

- **Land Temperature (1850–2025):** Gridded land surface air temperature anomaly record.
- **Sea Surface Temperature (1854–2025):** Ocean surface grid covering 70.9% Earth area.
- **Global Benchmark (1850–2025):** NOAA reference record for validation.
- **Atmospheric CO₂ (1979–2025):** Mauna Loa annual mean concentration (ppm).
- **Solar Irradiance (1850–2025):** Total Solar Irradiance (TSI in W/m²).
- **Volcanic Activity (1850–2025):** Volcanic Explosivity Index (VEI) & optical depth.

3. Surface Record Reconstruction

- Reconstructed series combines land (29.1%) and ocean (70.9%) across $N = 172$ years.
- **Linear Trend:** $+0.0586$ °C/decade ($R^2 = 0.6583$, $p = 1.71 \times 10^{-41}$).

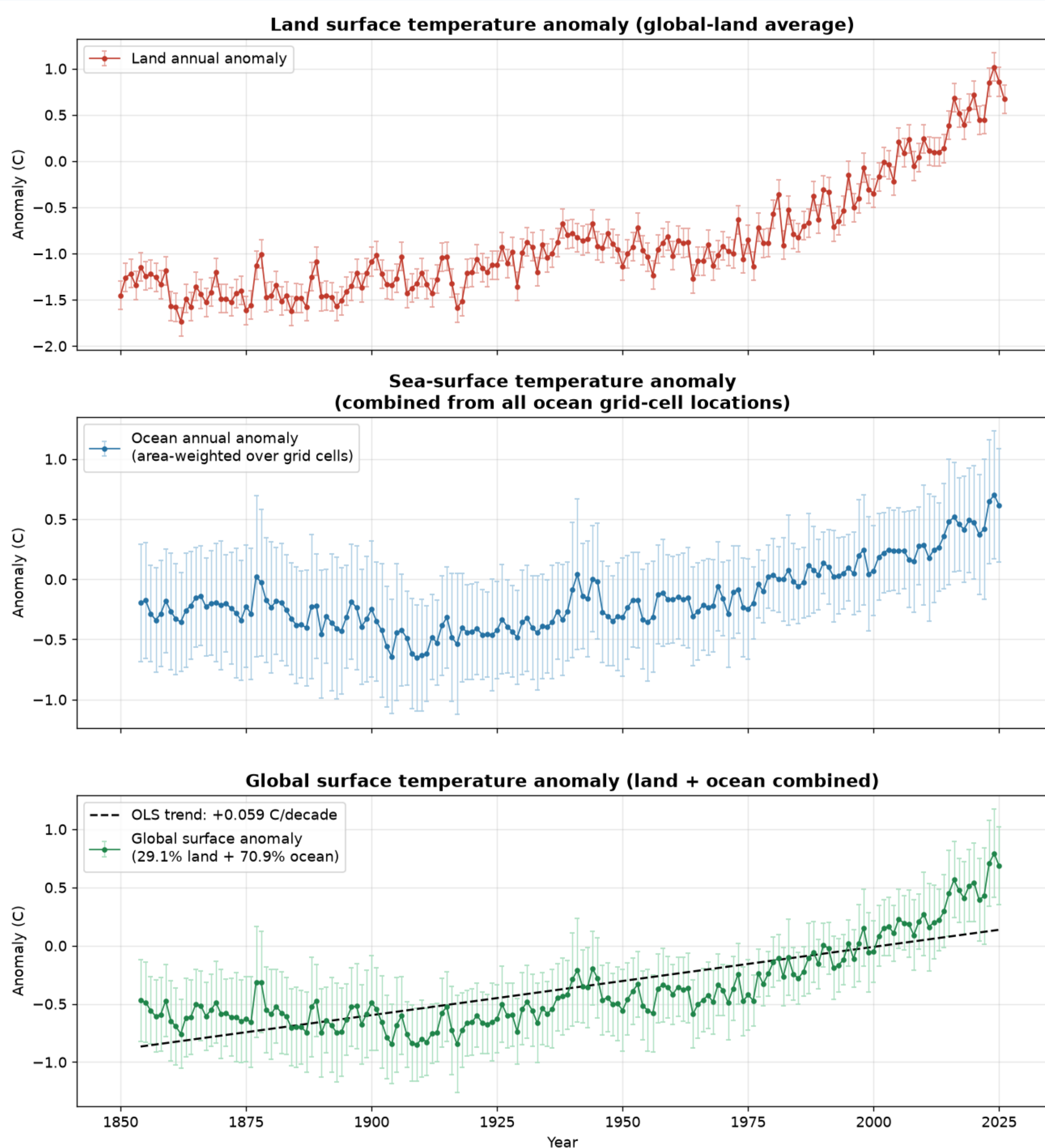


Figure 1: Reconstructed global temperature anomalies (1854–2025).

4. Benchmark Validation

- Custom record vs. NOAA reference shows strong agreement ($r = 0.998$).

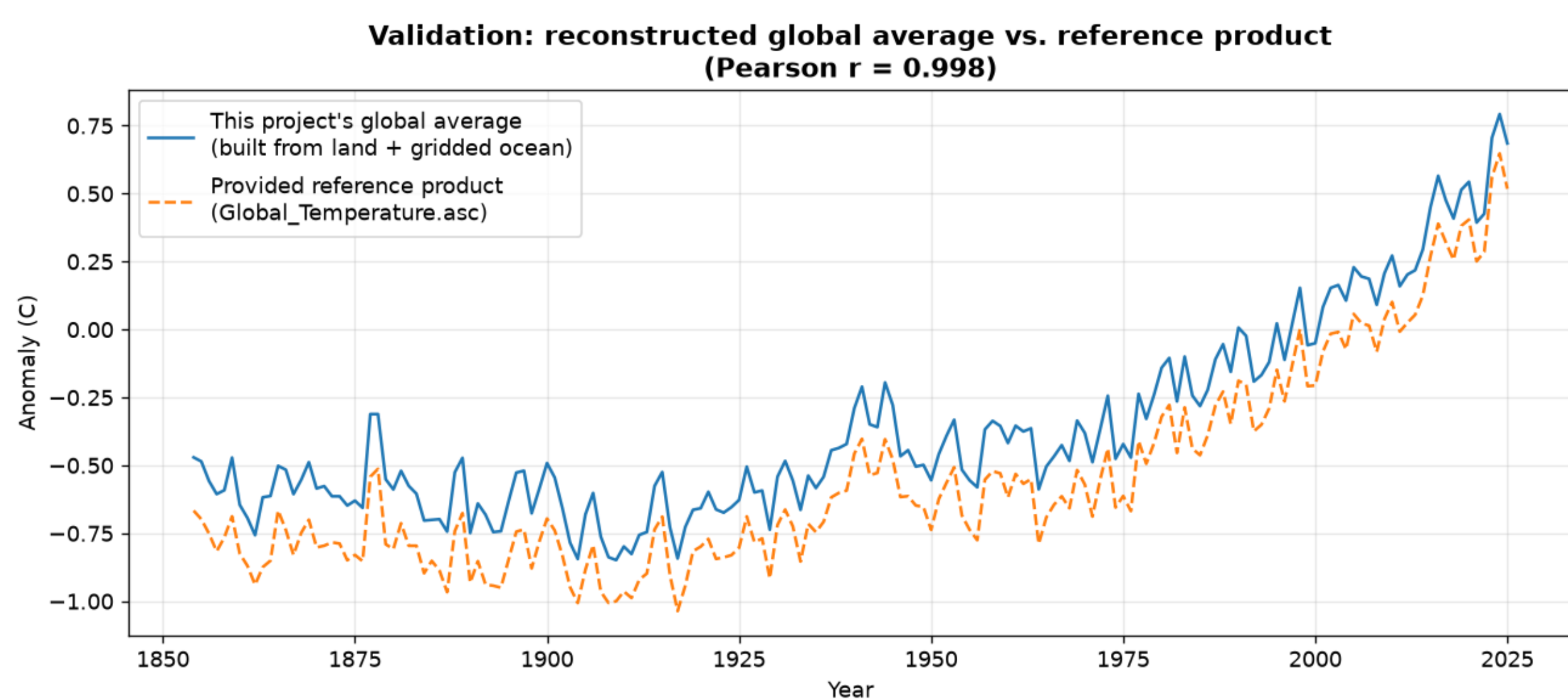


Figure 2: Verification of custom reconstruction against reference benchmark.

5. Pipeline Processing Workflow

- 1. Data Ingestion & QC**
Extract NetCDF/CSV datasets from raw files
- 2. Area Weighting**
 $\cos(\text{lat})$ ocean weighting (29.1% Land + 70.9% Ocean)
- 3. Autocorrelation Adjustment**
Bretherton N_{eff} estimation & first-differencing (ΔT)
- 4. Multivariate Attribution**
Standardized OLS regression with Newey-West HAC errors

6. Bivariate Drivers & Serial Adjustments

- **Unadjusted Correlation (1979–2025):** $r = 0.9500$, naive $p = 2.18 \times 10^{-24}$.
- **Bretherton Adjusted Correlation:** $N_{\text{eff}} = 5.1$, corrected $p = 0.0116$ (retains significance).
- **First-Difference Correlation (ΔCO_2 vs ΔT):** $r = 0.1771$, $p = 0.2391$.
- **Solar TSI Trend (1980–2025):** $+0.0027$ W/m²/dec ($p = 0.952$). Corrected TSI vs T : $r = 0.1422$, $p = 0.7127$.

7. Normalized Variable Trajectories

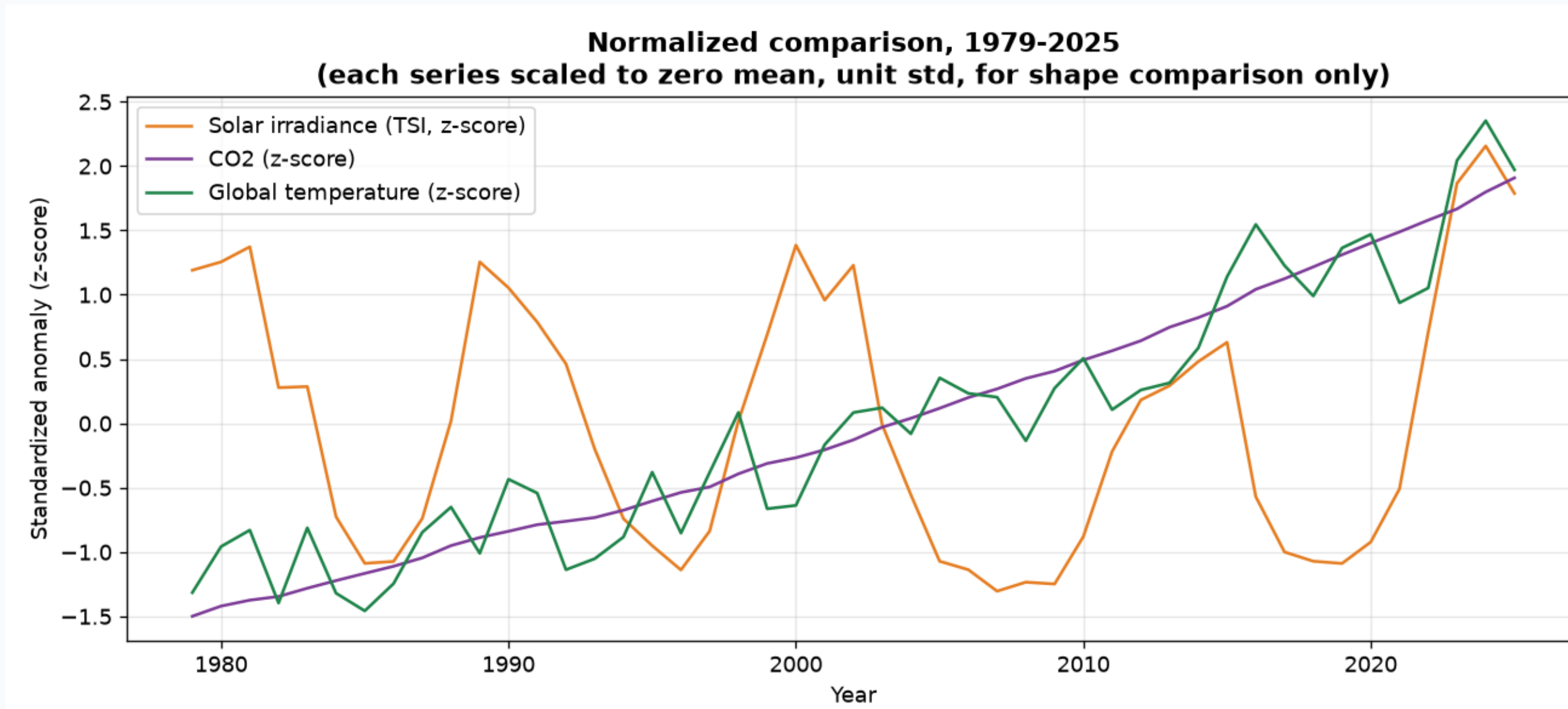


Figure 3: Standardized Z-scores for Solar TSI, Volcanic, and CO₂ (1979–2025).

8. Multivariate OLS Regression Results

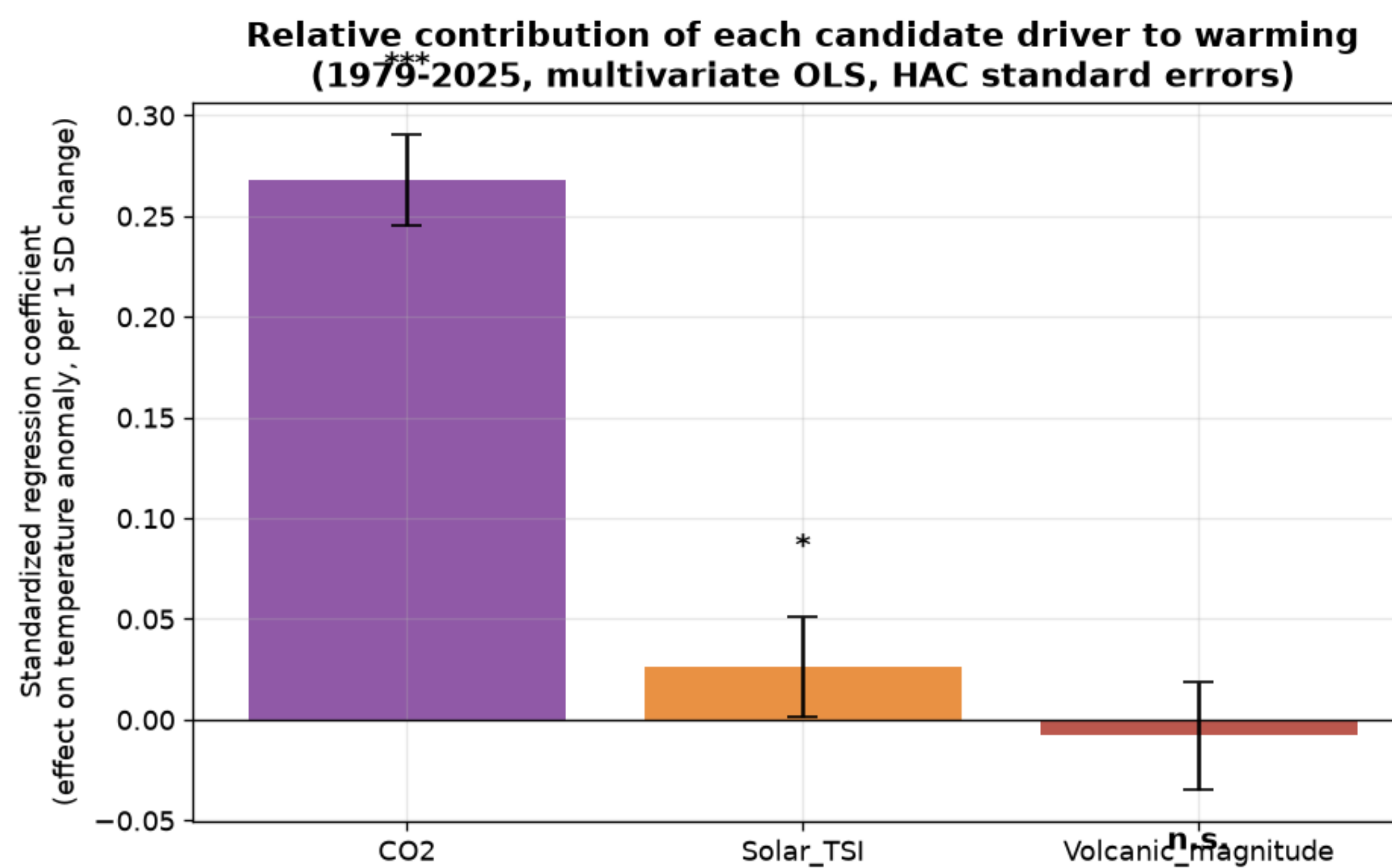


Figure 4: Standardized regression coefficients (β) with Newey-West HAC errors.

Predictor	Coef (β)	Std Err	z-statistic	$P > z $
Intercept	+0.1289	0.012	10.828	< 0.001
CO ₂	+0.2677	0.012	23.168	< 0.001
Solar TSI	+0.0263	0.013	2.079	0.038
Volcanic Mag	-0.0080	0.013	-0.596	0.551

Model Fit: $R^2 = 0.912$, Adj. $R^2 = 0.906$ ($F = 196.9$, $p < 0.001$, HAC 3-lag)

9. Key Conclusions

- **Dominant Anthro Forcing:** CO₂ is the primary driver of modern surface warming ($\beta = +0.2677$, $z = 23.168$, $p < 0.001$).
- **Scale Divergence:** CO₂ impact is $> 10\times$ greater than solar TSI variability ($\beta = +0.0263$, $p = 0.038$).
- **Insignificant Natural Forcings:** Solar TSI trends and volcanic eruptions do not account for post-1979 warming trends.



Scan for GitHub Code

Key References

1. Bretherton, C. S., et al. (1999). Effective spatial degrees of freedom. *J. Climate*, 12(7), 1990–2009.
2. Newey, W. K., & West, K. D. (1987). HAC Covariance Matrix. *Econometrica*, 55(3), 703–708.
3. NOAA & NASA NCEI (2025). Surface Temperature & Atmospheric CO₂ Records.